MILITARY STANDARD

MIL-STD 1320-22 (NAVY)

8 FEBRUARY 198[illegible]

TRUCKLOADING
VERTICAL LAUNCHING SYSTEM
MK 33 MOD 0 LOADER DOLLY
CANISTER/MISSILE

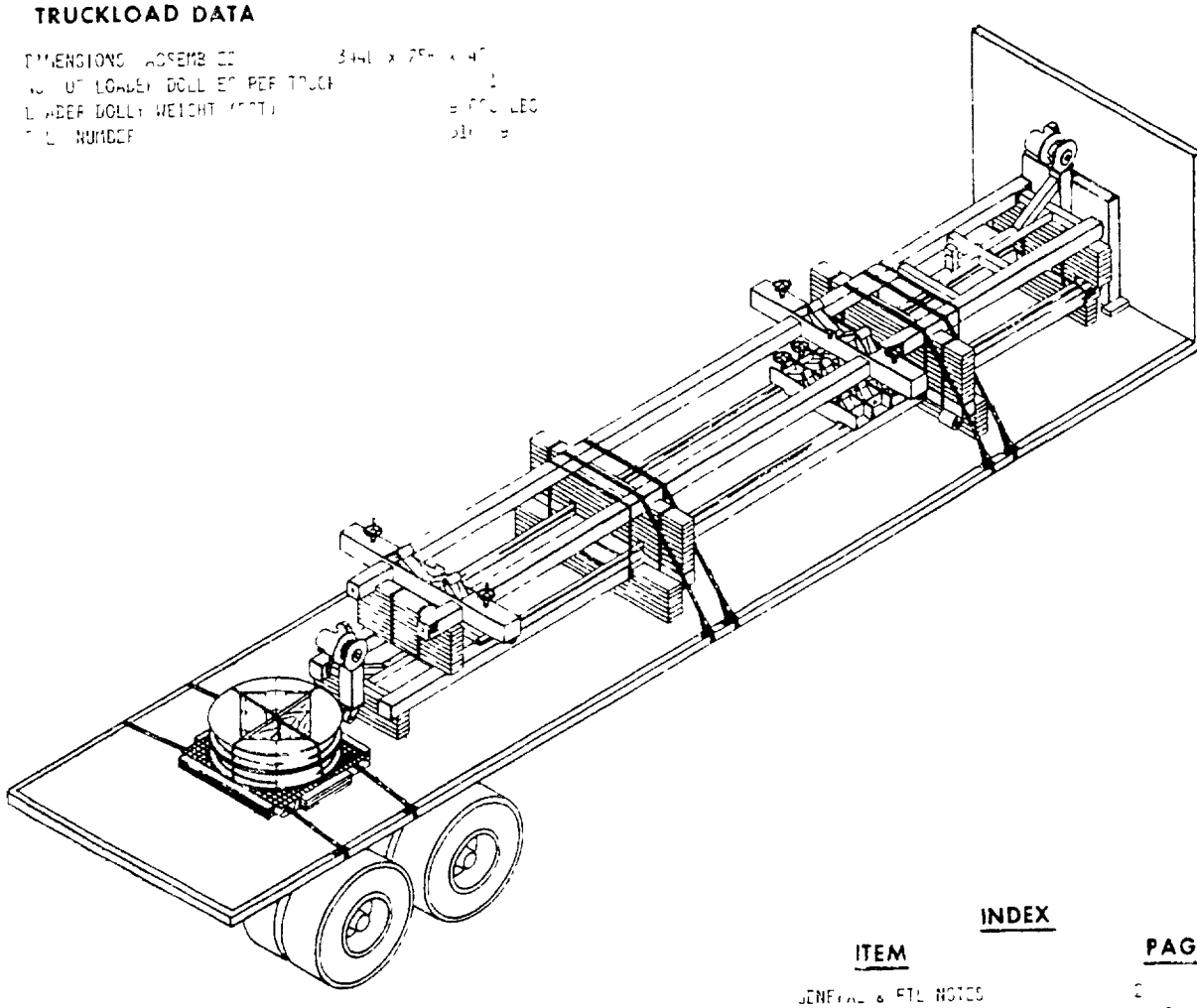
TRUCKLOAD DATA

DIMENSIONS ASSEMBLED [illegible] x [illegible] x [illegible]

NO. OF LOADER DOLLIES PER TRUCK [illegible]

LOADER DOLLY WEIGHT (LBS) [illegible] LBS

STOCK NUMBER [illegible]



ANSI NO. - N/A

INDEX

ITEM	PAGE
GENERAL & FTL NOTES	2
FTL	[illegible]
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LOADING	6

NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED TRUCKLOADING CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL HDBK 236 (NAVY)

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[illegible] NWHC 1 Feb 86
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

[illegible]
SIGNATURE CERTIFYING OFFICER

ORIGINATOR [illegible] 2/7/86
DATE

NAVAL WEAPONS SUPPORT CENTER, CRANE, INDIANA

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GENERAL AND FTL NOTES

AN FTL CONSISTS OF ONE COMPLETE CANISTER/MISSILE LOADER DOLLY. THIS DOCUMENT SHOWS ALL THE ACCESSORIES AND DETACHABLE COMPONENTS FOR THE SEVERAL CONFIGURATIONS OF THE LOADER DOLLY. SOME OF THESE ITEMS MAY BE MISSING FROM ANY ONE SHIPMENT. REFER TO THE APPROPRIATE OF-68 FOR COMPONENT SHIPPING DETAILS.

2. IT IS IMPORTANT THAT THE SEQUENCE OF EVENTS DESCRIBED IN STEPS 2 THROUGH 8 BE FOLLOWED PRECISELY.

3. WHEN PRESENT THE ROLL RINGS AND IDLER ASSEMBLIES ARE DETACHED FROM THE LOADER DOLLY AND PACKED SEPARATELY AS SHOWN IN DETAILS E AND F. THIS IS DONE PRIOR TO TRUCK LOADING.

4. PLACE PROTECTOR GATE ASSEMBLY, DETAIL D, AGAINST THE TRAILER'S PROTECTOR BOLTHEAD USING ITEMS 12 AND 13 AS SHOWN ON PAGE 6. ALSO LOCATE THE FORWARD SUPPORT ASSEMBLY AND NAIL AS SHOWN.

5. WHEN REQUIRED TO SHIP THE DRAW BAR AND "CHOPSTICK" ASSEMBLY, THE DRAW BAR IS APPLIED TO THE CHOPSTICKS AS SHOWN IN SECTION G-G PAGES 3 AND 4. PLACE THE TAPER ASSEMBLY WITH THE SMALLER DRAW BAR FITTING FORWARD OF AGAINST THE FORWARD SUPPORT AND CENTERED ON THE WIDTH OF THE TRAILER.

6. NAIL REMAINING SUPPORT ASSEMBLIES (MID FRONT WITH NOTCH, MID REAR AND AFT END) AS PALLET FRONT BLOCKING AS SHOWN ON PAGE 4. THE NOTCH IN THE MID FRONT ASSEMBLY SHOULD STRADDLE THE DRAW BAR CHOPSTICK ASSEMBLY. NAIL THE NOTCHED RESTRAINT ITEMS 2E AND 2F AS SHOWN ON PAGE 5. ALSO NAIL THE ACCESSORY SUPPORT CLEATS, ITEMS 2G AND 2H AS SHOWN.

7. SECURE MISSILE LOADER CONNECTING LATCHES CLOSED WITH GLASS REINFORCED TAPE, ITEM 21. LOCK THE MAIN CASTERS AND THEN STRAP THE CANISTER LOADER SUPPORT ASSEMBLIES, DETAILS A, B, AND C TO THE MISSILE LOADER WITH 1 1/4 INCH STRAP, ITEM 22 AS SHOWN ON PAGES 5 AND 6.

8. USING TWO FORKLIFTS, PLACE THE MISSILE LOADER DOLLY WITH THE MINOR END AFT ONTO THE FOUR SUPPORT ASSEMBLIES SO THAT THE CASTER BEAMS ARE INSIDE THE MID FRONT AND MID REAR ASSEMBLIES AND THE LONGITUDINAL RAILS SIT IN BETWEEN THE END BLOCKING OF THESE ASSEMBLIES AS SHOWN ON PAGE 3.

9. STRAP THE TWO IDLER SUPPORTS AND THE SPACER ASSEMBLY TOGETHER AS SHOWN IN DETAIL E. THEN STRAP THE SUPPORTS AND SPACER TO ACCESSORY BOX A1 WITH TWO STRAPS, ITEM 5. PLACE USING A SUITABLE SLING PLACE THE ACCESSORY BOX AND IDLER SUPPORTS BETWEEN THE RAILS OF THE MISSILE LOADER ON TOP OF THE CLEATS AND UP AGAINST THE SUPPORT HOUSINGS AS SHOWN ON PAGE 7. PLACE ENTABLE TO ITEM 16, PAGE 8 AS SHOWN AND NAIL TO THE TRAILER FLOOR.

10. LOAD THE CANISTER LOADER ONTO ITS SUPPORT ASSEMBLIES SO THAT THE MINOR SUPPORT POST BEARS AGAINST THE PROTECTOR GATE ASSEMBLY. USE TWO FORKLIFTS.

11. POSITION THE FOUR UNITIZING STRAPS, ITEM 20. TENSION AND DOUBLE NOTCH SEAL.

12. IF PRESENT, PLACE ACCESSORY BOXES 2 AND 3 WITHIN THE SIDE RAILS OF THE TWO LOADER DOLLY COMPONENTS AND ON TOP OF THE CLEATS (WHERE APPLICABLE) AS SHOWN ON PAGES 6 AND 7. LOCATE THE BOXES AGAINST LEG BEAMS AND SECURE WITH DOUBLE 2 x 4 x 18 FLOOR BLOCKING AS REQ'D. FLOOR BLOCKING IS NOT DETAILED BECAUSE ACCESSORY BOX DESIGN IS PENDING.

13. INSTALL TIE DOWN STRAP, ITEM 23 TENSION AND SEAL. CHAINS AND LOAD BINDERS ARE AN ACCEPTABLE SUBSTITUTE IF SUITABLE PROTECTORBOARD (DOUBLED 2 x 6 OR APPROVED NYLON WEBBING TIEDOWNS ARE ALSO ACCEPTABLE.

14. PALLETIZE THE ROLL RINGS AS SHOWN IN DETAIL F. LOAD PALLET WITH TYPE 4 NOSE SIDE UP AGAINST THE PALLET FRONT ASSEMBLY ITEM 4. INSTALL REMAINING PALLET BLOCKING ITEMS 4 AND 5, AND NAIL AS SHOWN. THREAD TIE DOWN STRAPS THROUGH PALLET LEGS AS SHOWN. TENSION AND SEAL.

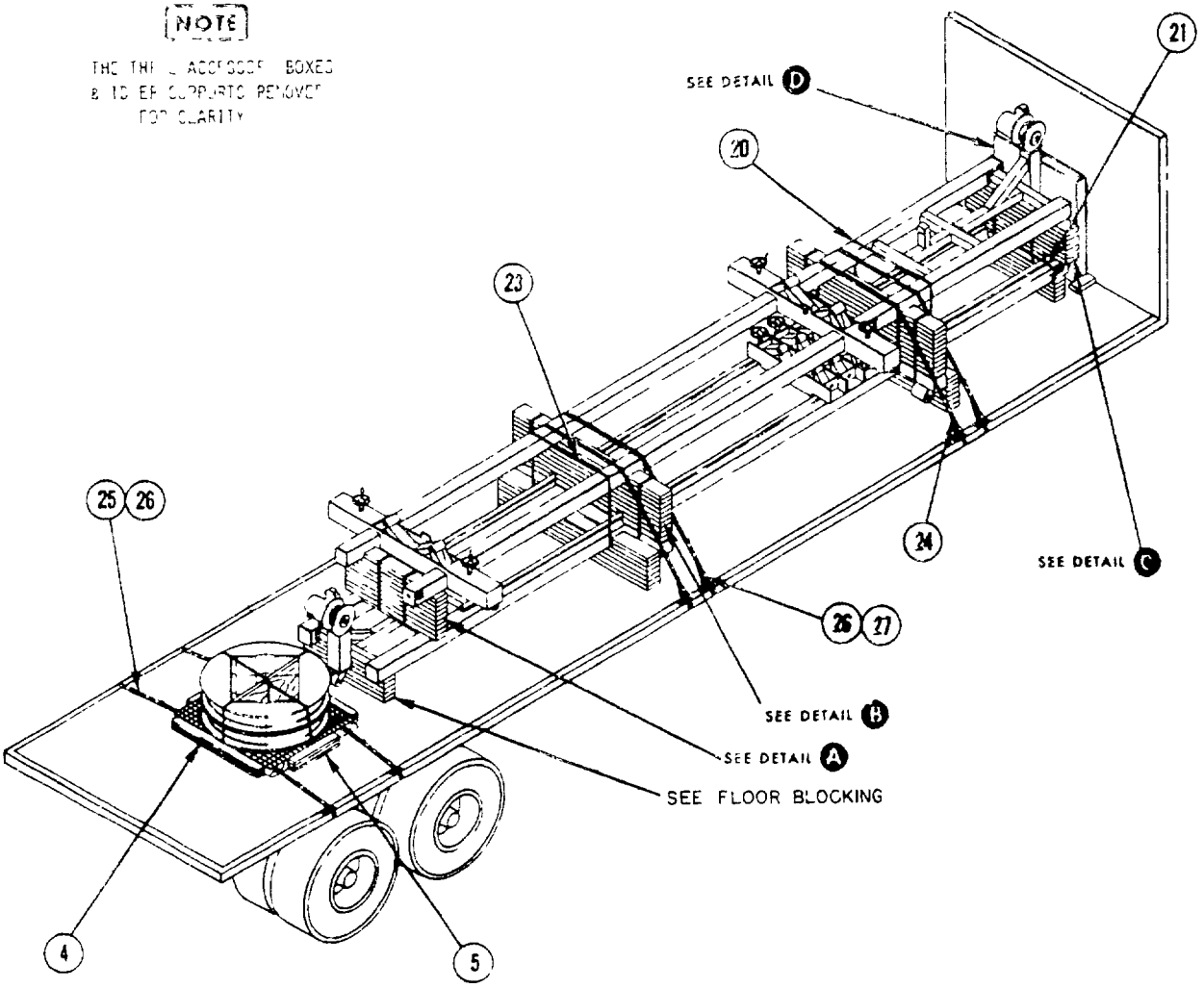
15. RECEIVING ACTIVITIES SHOULD CAREFULLY REMOVE AND SAVE THE LAMINATED WOOD CANISTER LOADER SUPPORTS, THE FRONT PROTECTOR GATE, AND THE NOTCHED DRAW BAR RESTRAINTS FOR RE-USE.

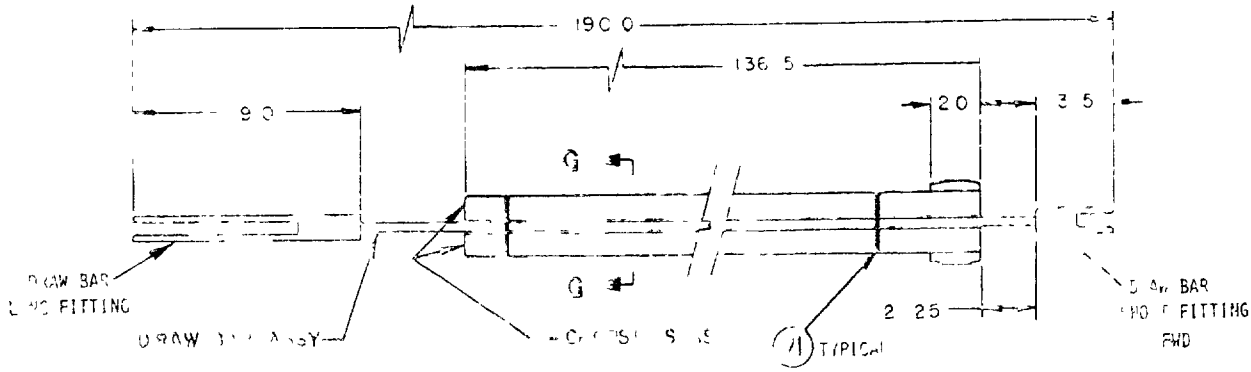
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FTL 40, 42, 44, AND 45 FT TRAILERS (FLATBED)

NOTE

THE THREE ACCESSORY BOXES & TIE DOWN SUPPORTS REMOVED FOR CLARITY





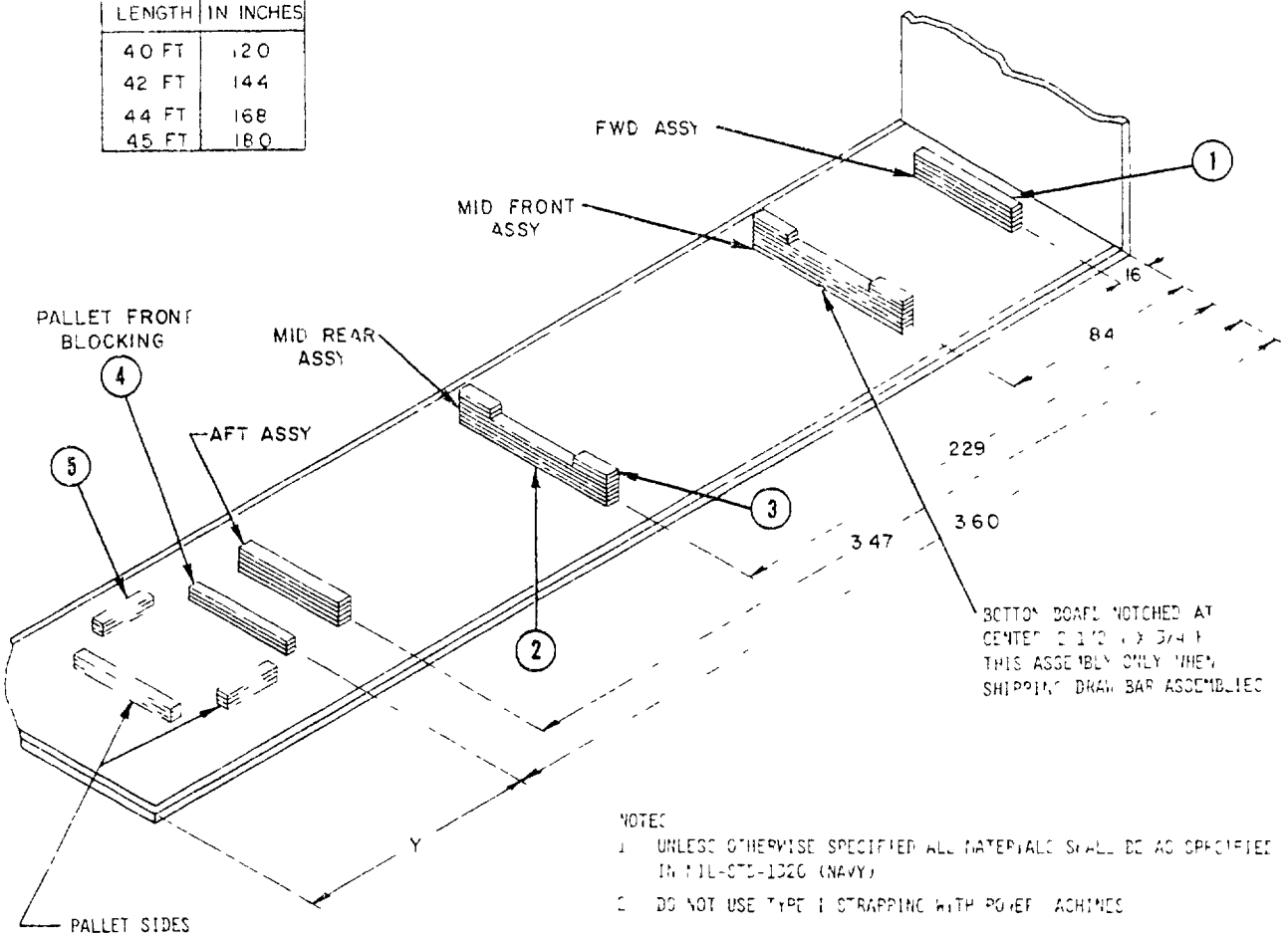
DRAW BAR AND CHOPSTICKS ASSY

1. FAPE DRAW BAR & CHOPSTICKS W/ PLASTIC TAPE ... SECTION C-S APPROX VERY 24 IN

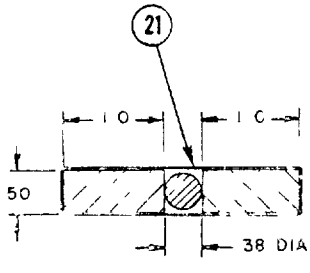
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FLOOR BLOCKING

TRAILER LENGTH	DIM "Y" IN INCHES
40 FT	120
42 FT	144
44 FT	168
45 FT	180



NOTES:
1. UNLESS OTHERWISE SPECIFIED ALL MATERIALS SHALL BE AS SPECIFIED IN MIL-STD-1320 (NAVY)
2. DO NOT USE TYPE I STRAPPING WITH POWER MACHINES

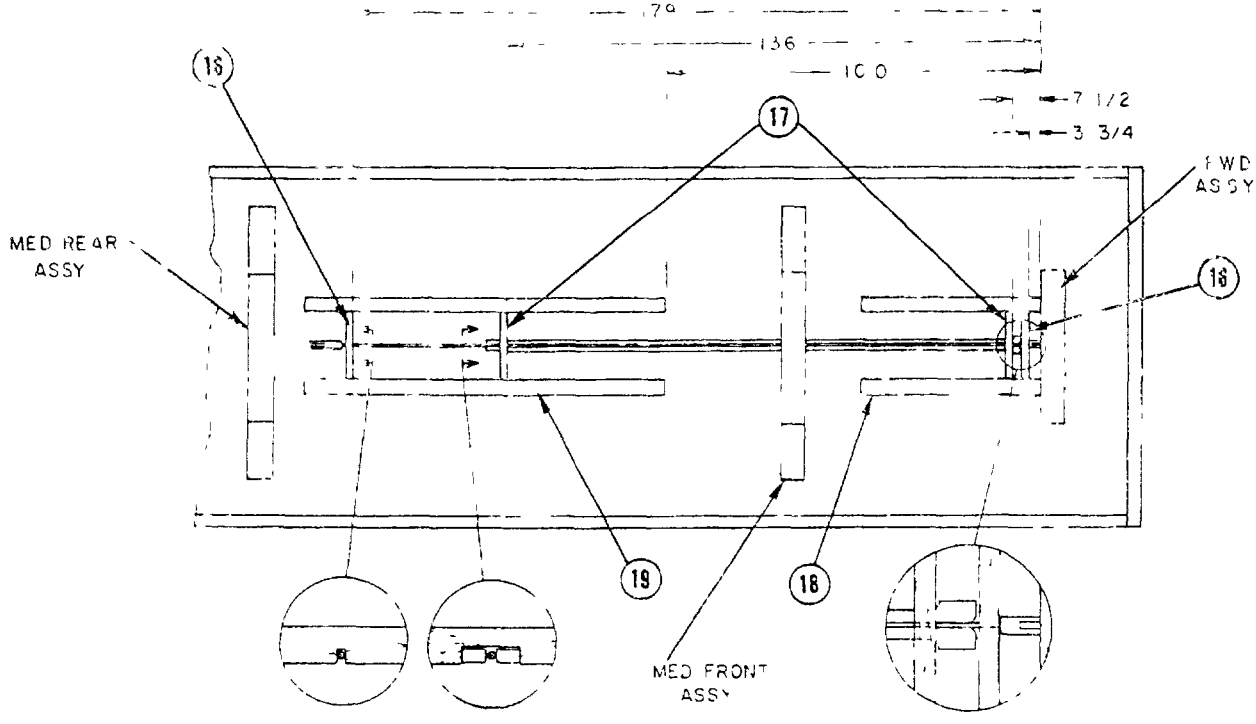


SECTION G-G

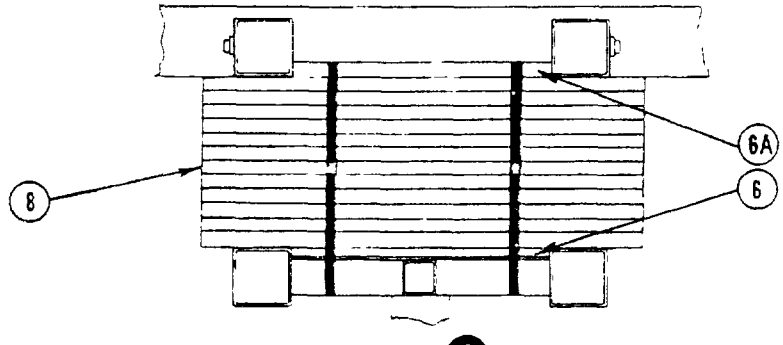
REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
2	12	FRONT PROTECTOR BLOCKING	SEE NOTE 1	2 X 4 X [illegible]
2	11	FRONT PROTECTOR BLOCKING	SEE NOTE 1	2 X 4 X 30
20	10	MID CANISTER SUPPORT	SEE NOTE 1	[illegible]
4	10	AFT CANISTER SUPPORT TOP	SEE NOTE 1	2 X 6 X 36
8	9	MID CANISTER SUPPORT BOTTOM	SEE NOTE 1	2 X 6 X 27-1/4
22	8	FWD OR AFT CANISTER SUPPORT	SEE NOTE 1	2 6 48
4	7	MID CANISTER SUPPORT BOTTOM	SEE NOTE 1	2 X 6 X 3
1	6	CANISTER SUPPORT TOP	SEE NOTE 1	2 X 6 27-3/4
2	6	CANISTER SUPPORT BOTTOM	SEE NOTE 1	2 X 6 X 27-3/4
6	5	PALLET SIDE BLOCKING	SEE NOTE 1	2 X 4 X 24
6	4	PALLET FRONT OR REAR BLOCKING	SEE NOTE 1	2 X 4 X 40
8	3	MID FRONT OR MID REAR END BLOCKS	SEE NOTE 1	2 X 6 X 16-1/4
10	2	MID FRONT OR MID REAR SUPPORT	SEE NOTE 1	2 X 6 X 73
10	1	FWD OR AFT SUPPORT	SEE NOTE 1	2 X 6 X 40

LIST OF MATERIALS

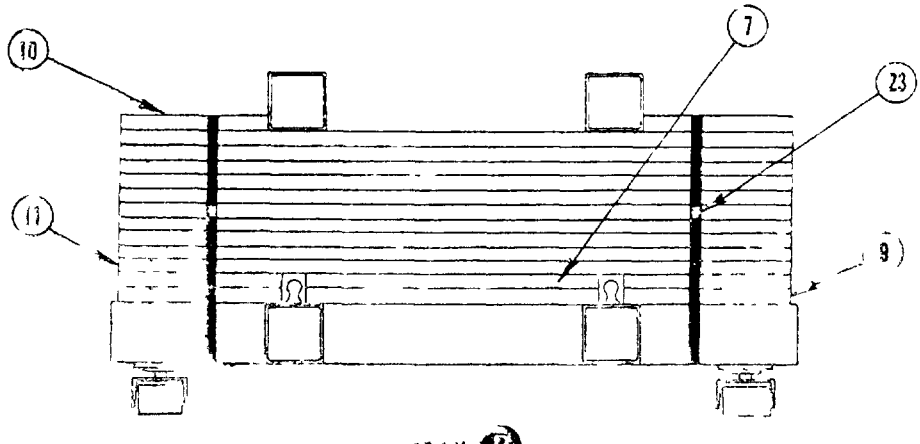
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FLOOR BLOCKING FOR DRAW BAR & CHOPSTICKS ASSY

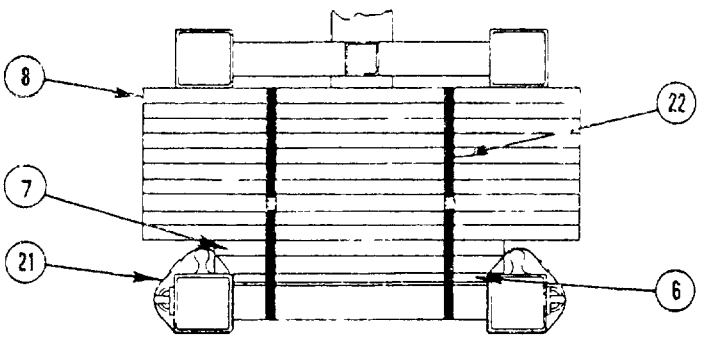


DETAIL A

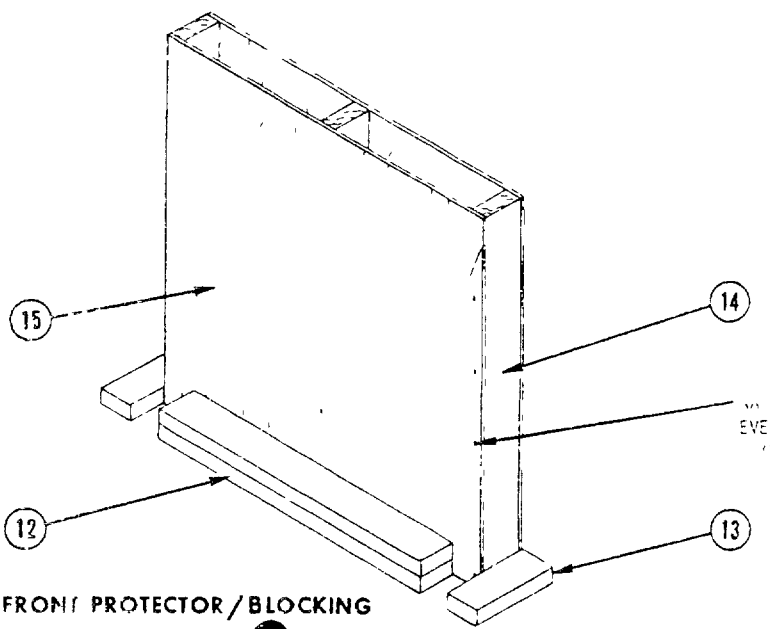


DETAIL B
(2 PLACES)

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DETAIL C

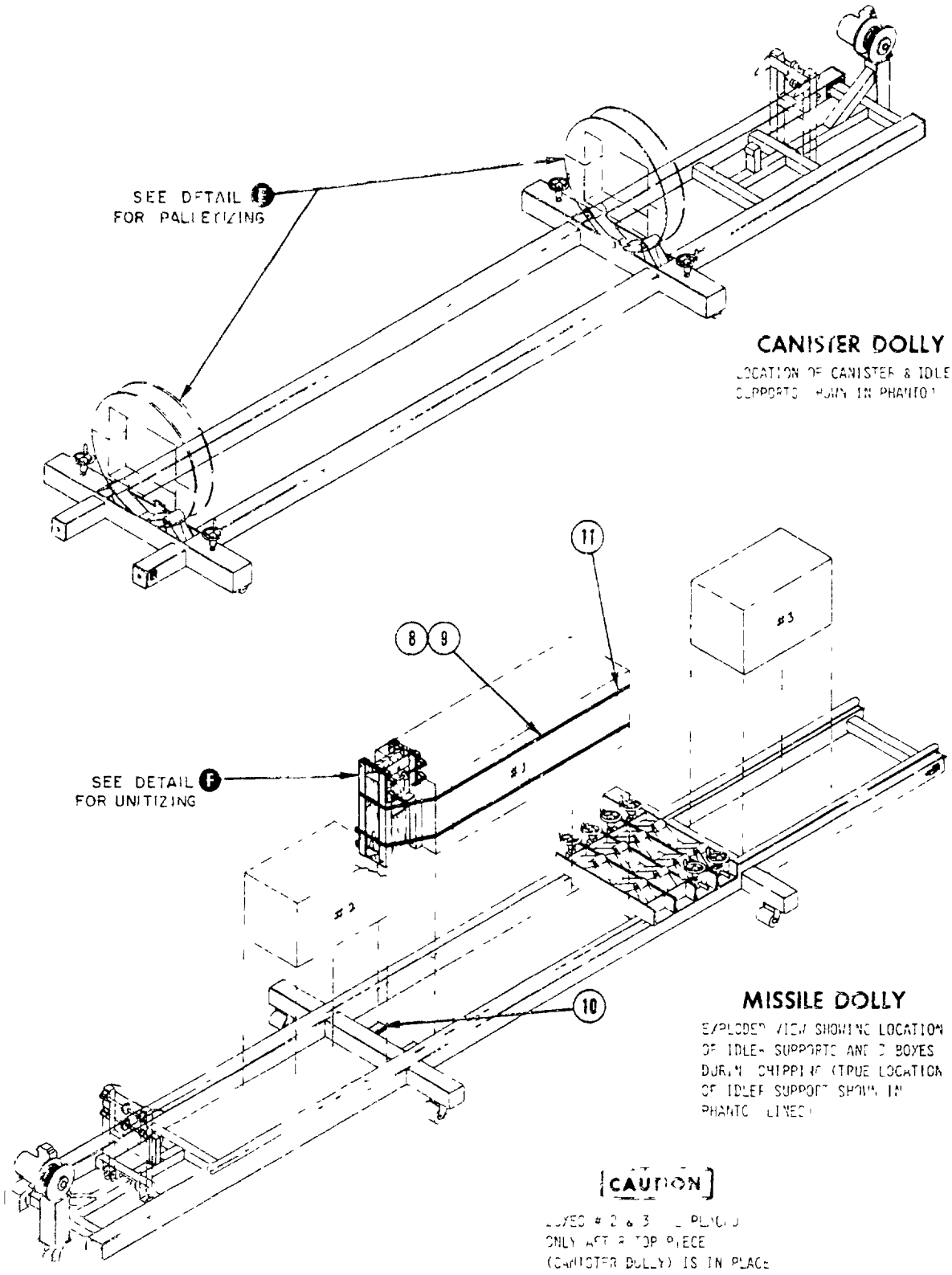


FRONT PROTECTOR/BLOCKING
DETAIL D

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
12	27	STAKE POCKET (WITH PIN)	SEE NOTE 1	2 [illegible]
48	26	SEALS	SEE NOTE 1	[illegible]
2	25	PULLEY TIE DOWN STRAPS	SEE NOTE 1	2 [illegible]
4	24	LOADED ROLL TIE DOWN STRAPS	SEE NOTE 1	2 x 600 [illegible]
12	23	SEALS	SEE NOTE 1	[illegible]
8	22	CANISTER SUPPORT STRAPS	SEE NOTES 1 & 8	[illegible]
2	21	LATCH RESTRAINT BLOCK TYPE	SEE NOTE 1	[illegible]
4	20	MATERIAL STRAPS	SEE NOTES 1 & 2	[illegible]
2	19	ACCESSORY SUPPORT CLEATS	SEE NOTE 1	[illegible]
2	18	ACCESSORY SUPPORT CLEATS	SEE NOTE 1	[illegible]
2	17	NOTCHED DRAW BAR RESTRAINT	2-1/2X3/4H NOTCH	2 x 2 x 12
2	16	NOTCHED DRAW BAR RESTRAINT	3/8X3/8H NOTCH	2 x 2 x 18
2	15	FRONT PROTECTOR PANEL	PLYWOOD	1/2 X 35 X 50
3	14	FRONT PROTECTOR BATTEN	SEE NOTE 1	2 x 4 x 36

LIST OF MATERIALS

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CANISTER DOLLY

LOCATION OF CANISTER & IDLER SUPPORTS SHOWN IN PHANTOM

MISSILE DOLLY

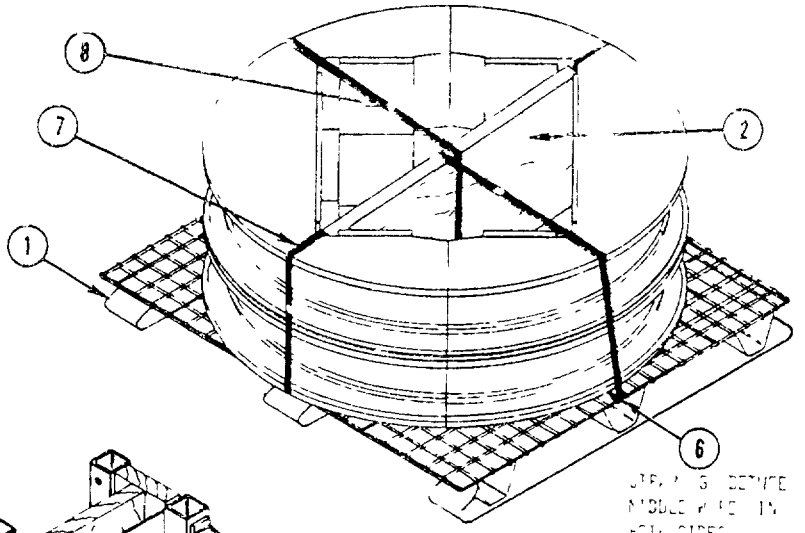
EXPLODED VIEW SHOWING LOCATION OF IDLER SUPPORTS AND 3 BOXES DURING SHIPPING (TRUE LOCATION OF IDLER SUPPORT SHOWN IN PHANTOM LINES)

CAUTION

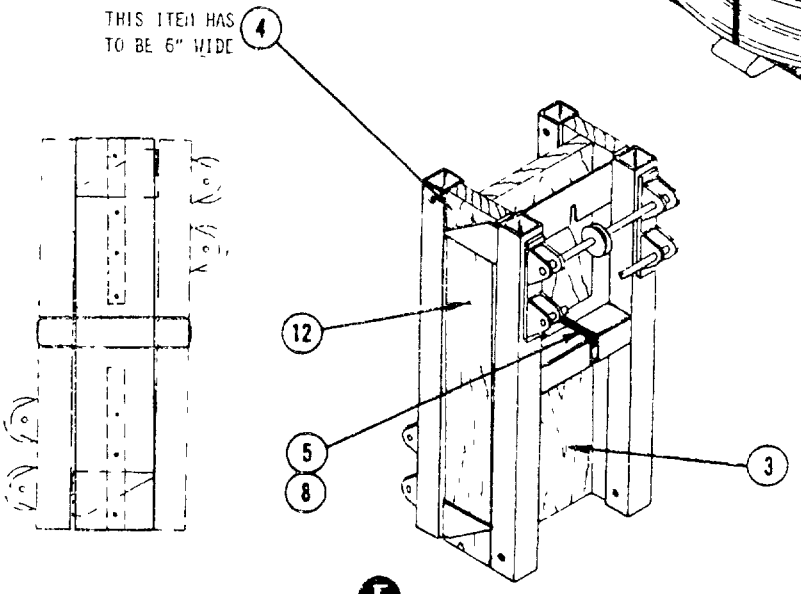
BOXES # 2 & 3 [illegible] PLACED ONLY AFT OF TOP PIECE (CANISTER DOLLY) IS IN PLACE

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UNITIZING PROCEDURE



DETAIL E



DETAIL F

NOTES

1 UNLESS OTHERWISE SPECIFIED ALL MATERIAL SHALL BE [illegible] IN MIL-STD-1320 (NAVY)

2 DO NOT USE TYPE 1 STRAPPING WITH POWER SEALED [illegible]

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
16	14	NAIL, COMMON	SEE NOTE 2	10d 4/JOINT
AR	11	STAPLES	COMM'L	5/4
4	10	END CLEAT	SEE NOTE 2	2 x 4 x 18
2	9	ACCESSORY BOX / RAP	SEE NOTE 2 & 3	5/4 .03" x 39 F
7	8	SEALS	SEE NOTE 2	5/4
2	7	PALLET STRAP	SEE NOTE 1 & 2	3/4 x .035 4 FT
2	6	PALLET STRAP	SEE NOTE 1 & 2	3/4 x .035 7 FT
2	5	IDLER SUPPORT STRAP	SEE NOTE 1 & 2	3/4 x .03" 3 FT
2	4	IDLER SUPPORT BATTEN	* ACTUAL DIM REQ'D SEE FIELD NOTE	2 x 4 x 35
2	3	IDLER SUPPORT BATTEN	SEE NOTE 2	2 x 12 x 15
1	2	ROLL RING SPACER	SEE NOTE 1	2 x 12 x 36
1	1	PALLET, MK 3 MOD 0	DWG 564200	40 x 48 x 4 3/4

LIST OF MATERIALS

REVIEW ACTIVITY
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
(PROJECT NO 8140 - N671)

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(See Instructions - Type or Print)

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MIL-STD-1320-224 (NAVY)

2. DOCUMENT TITLE

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b. ADDRESS (*Street, City, State, ZIP Code*)

4. TYPE OF ORGANIZATION (*Mark one*)

- [ ] VENDOR
- [ ] USER
- [ ] MANUFACTURER
- [ ] OTHER (*Specify*)

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER (*Last, First, MI*) — Optional

b. WORK TELEPHONE NUMBER (*Include Area Code*) — Optional

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